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Evaluating Encoding Strategies and Quantum Circuit Designs for Sentiment Classification

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***Abstract*—We present a component-wise ablation study of a quantum natural language processing (QNLP) pipeline for sentiment classification, focusing on how classical and quantum design choices interact under NISQ constraints. Starting from a baseline, we evaluate five different qubit numbers, six sentence embedding strategies, two dimensionality reduction methods, two feature encoding schemes, and three variational ansatzes. Each stage is tested independently while fixing previously optimized components, allowing us to isolate the contribution of each. Experiments are conducted on simulated quantum backends. We find that sentence-level transformer embeddings (MiniLM, MPNet), linear dimensionality reduction (PCA), and simple angle encoding provide substantial accuracy gains without additional training cost. Finally, we introduce a shallow custom ansatz that induces a multiplicative cosine kernel across qubits. While the best overall test accuracy (87%) is achieved using the RealAmplitude ansatz, the custom ansatz closely follows (86.5%) while reducing runtime by over 35% (277 s vs. 432 s). This demonstrates a promising trade-off between performance and efficiency, supporting the design of scalable QNLP pipelines under limited quantum resources.**

***Index Terms*—Quantum natural language processing, variational quantum circuits, sentiment analysis, ansatz design**

I. Introduction

Quantum computing offers the potential to solve certain computational problems exponentially faster than classical approaches, leveraging principles such as superposition and entanglement [1]. Speedups have been demonstrated for tasks ranging from unstructured search to structured algebraic problems like the hidden subgroup problem [2]. However, current quantum hardware, referred to as Noisy Intermediate-Scale Quantum (NISQ) devices [3], is fundamentally constrained by limited qubit counts and decoherence, motivating the development of algorithms that are both expressive and resource-efficient.

Quantum Natural Language Processing (QNLP) is a growing field that explores how quantum algorithms can be applied to model and process human language [4]. Although still in its early stages, several studies have demonstrated QNLP's potential in a variety of NLP tasks. One particularly relevant area is sentiment analysis, which leverages NLP to infer sentiment embedded in text. Sentiment analysis has high practical value, allowing businesses and public institutions to monitor opinions, manage reputation, and support decision-making [5].

Sentiment analysis is one of the most widely studied tasks in natural language processing [6]. It aims to determine the emotional tone conveyed in a text, often by classifying it as positive, negative, or neutral. Common applications include analyzing customer reviews, social media posts, and survey responses. By training on labeled examples, sentiment analysis systems can automatically detect the writer's attitude or opinion, making it a valuable tool in industry and research [7].

A widely used framework in QNLP is the DisCoCat (Categorical Compositional Distributional) model, which maps grammatical structures into string diagrams. These diagrams are then compiled into quantum circuits, allowing sentence meanings to be composed via the structure of quantum operations [8].

To make QNLP practically executable, Kartsaklis *et al.* [9] introduced the *lambeq* toolkit, which automates the conversion of natural language into string diagrams and then into parameterized quantum circuits for hybrid training. This pipeline has been successfully applied to sentiment classification tasks. Early results from Martinez *et al.* [10] and Ganguly *et al.* [11] demonstrate the feasibility of using QNLP for sentiment classification, achieving high accuracy on small, curated datasets.

However, the scalability of these approaches remains a key challenge. String diagram-based pipelines often allocate one qubit per token and produce bespoke quantum circuits per sentence, which becomes intractable on NISQ devices [12]. Addressing this bottleneck is critical to increase the capacity of QNLP-based sentiment classifiers to process longer, more varied language input drawn from real datasets.

One approach to address the bottleneck is done by Masum, who applied variational quantum circuits (VQCs) to sentiment analysis using a Bag-of-Words encoding scheme combined with a hardware-efficient ansatz based on RY gates [13]. The study showed that with the right combination of encoding and ansatz design, it is possible to reduce the qubit requirements. This

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finding underscores the strong influence that encoding and ansatz choices have on qubit efficiency.

Two crucial components in this framework are the sentence encoding strategy and the quantum ansatz design. Compact encodings must reduce redundancy and minimize qubit usage, while the ansatz must be expressive, trainable, and compatible with hardware constraints. The joint optimization of these two components is key to building scalable QNLP systems in the NISQ era.

Despite prior efforts, there is limited empirical analysis that systematically explores the interplay between encoding strategies and quantum circuit structures in QNLP tasks. Existing studies typically focus on one or two configurations, making it difficult to assess trade-offs between accuracy, training time, and qubit efficiency across a broader design space. A more comprehensive evaluation is needed to guide practical circuit and encoding design under hardware constraints.

This motivates the central question of this paper, which is: how do different sentence encoding strategies and quantum circuit designs impact the performance and scalability of QNLP-based sentiment classifiers on NISQ-era devices?

To answer this, we conduct a modular evaluation of encoding strategies and quantum circuit configurations for sentiment classification. We perform a systematic ablation study across combinations of sentence encodings and quantum circuit designs, analyzing trade-offs in accuracy, resource usage, and training time. Additionally, we introduce and evaluate a custom ansatz design optimized for low-qubit settings.

II. Related Work

Early empirical evidence that quantum circuits can handle sentiment comes from two small-scale demonstrations. Masum *et al.* compress Bag-of-Words vectors with PCA (2–6 dimensions), encode the features with a second-order Pauli feature map, and classify them using a six-qubit Variational Quantum Circuit based on the Real Amplitude ansatz with linear entanglement despite the small qubit count, the model retains approximately 71–72% accuracy while training significantly faster than a classical SVM on the same datasets of 1 619 Bengali news snippets and 728 English tweets [13]. In contrast, Ganguly *et al.* feed 130 labeled sentences through the `lambeq` pipeline to obtain string diagram-based IQP-style circuits. These circuits are run on four-qubit hardware with perfect test accuracy in noiseless simulation and competitive performance on IBM-Q emulators, showcasing syntax-aware compilation strategies [11]. Together, these studies suggest that QNLP can achieve competitive accuracy on curated sentiment tasks, albeit with limitations in scalability and generalization due to sentence-specific circuits.

A large body of classical work addresses sentence encoding and dimension reduction before classification. FastText, which encodes sentences as bag-of-character n-grams, trains a linear classifier on a billion-token corpus in under 10 minutes while matching deep CNN/LSTM accuracy on 14 public benchmarks [14]. Word2Vec-based representations such as WARV (Weighted Average of Re-weighted Vectors) further improve performance across IMDB, Yelp, and SemEval with minimal computational overhead [15]. Sentence transformers offer higher accuracy but at a higher embedding cost: in document retrieval, MiniLM encodes over 6,000 abstracts faster than MPNet, but the latter yields significantly better median ranks, highlighting a speed–accuracy trade-off [16]. Dimensionality reduction is essential for qubit feasibility: Masum *et al.* employ PCA (with optional Haar transforms), while Beaulieu *et al.* visualize 2D clusters from quantum reservoir states using UMAP [17], reinforcing the importance of compact input features in quantum NLP.

Mapping classical data into quantum states is a design decision that significantly impacts performance and resource use. Pauli feature maps, as used by Masum *et al.*, encode input vectors through products of Pauli operators, creating expressive high-order kernels suitable for quantum SVMs and VQCs [13]. This method builds on the Pauli-Z maps proposed by Havlíček *et al.*, which are provably hard to simulate classically and provide strong kernel expressivity [18]. In contrast, angle encoding, where classical features are mapped to rotation angles, remains widely used due to its hardware efficiency and constant-depth circuits, though it trades off representational power for simplicity [19].

Once data is encoded, variational circuits (ansatz) shape the learning capacity of quantum models. Masum *et al.* employ the Real Amplitude ansatz with linear entanglement, showing that even shallow structures can reach competitive performance under qubit constraints [13]. The Strongly Entangling Layer (SEL) ansatz, proposed by Schuld *et al.*, offers enhanced expressivity through cyclic entanglement and parameterized single-qubit layers, and is now a standard in platforms like PennyLane due to its performance in noisy environments [20]. Choosing the appropriate ansatz involves weighing circuit depth, expressivity, and hardware limitations.

III. Methodology and Experiment Setups

A. Experimental Setup

This experiment uses a general pipeline for quantum-enhanced sentiment classification, as illustrated in Fig. 1. The pipeline consists of four major stages: sentence encoding, dimensionality reduction, quantum circuit construction, and quantum circuit training. Each configuration is evaluated to measure classification accuracy and fit time.

The experimental pipeline builds on the structure proposed by Masum *et al.*, who combined a Bag-of-Words encoder with PCA, Pauli encoding, and a Real-Amplitude ansatz [13]. This configuration serves as our baseline. All configurations are evaluated across three random seeds. A staged, component-wise optimization is then conducted: in each step, one component is

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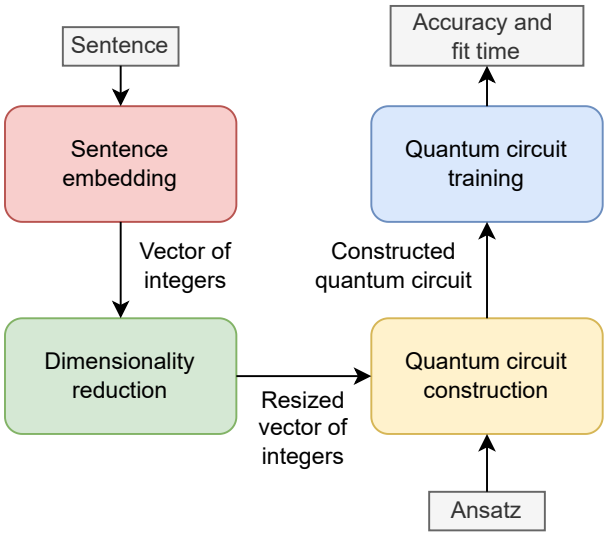


Fig. 1: Custom QNLP Pipeline for Sentiment Classification

varied while others are fixed to their current best-performing setting, allowing us to isolate the contribution of each design choice.

We begin by varying the number of qubits $k \in \{2, 3, 4, 5, 6\}$, since circuit width and fidelity are directly affected by qubit count. With k fixed, we then compare six sentence embedding methods: (i) Bag-of-Words (baseline) and TF-IDF as sparse vector encoders; (ii) Word2Vec and FastText as word-level embedding approaches; and (iii) MiniLM and MPNet as dense sentence-level encoders.

Next, the high-dimensional embeddings are compressed using either Principal Component Analysis (PCA), the baseline linear method, or Uniform Manifold Approximation and Projection (UMAP), a non-linear alternative. The reduced vectors are then mapped into quantum states via one of two encoding schemes, Pauli encoding (baseline), where features modulate rotations of Pauli operators, and Angle encoding, a simple, non-entangling alternative that directly uses feature values as rotation angles.

Finally, we assess three variational circuit designs, the RealAmplitude ansatz, the Strongly Entangling Layer (SEL), and a custom cosine-interference ansatz combining Hadamard layers, R_y rotations, linear entanglement via CZ gates (Fig. 2). Each configuration is trained and evaluated using accuracy metrics across train, validation, and test splits, averaged over three seeds to ensure stability and reproducibility.

B. Custom Cosine-Interference Ansatz

The circuit in Fig. 2 is designed so that the expectation value of the read-out qubit collapses to a global product of cosines. Each sentence feature x_j is first encoded by an $R_y(x_j)$ rotation, followed by a Hadamard gate that spreads amplitude across the

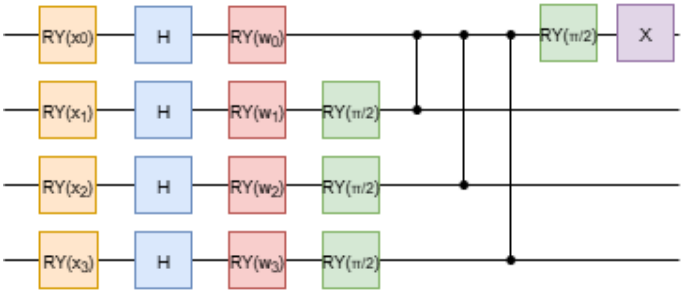


Fig. 2: Configuration of Custom Ansatz

computational basis. A trainable rotation $R_y(w_j)$ introduces variational parameters, and a ring of controlled-Z (CZ) gates entangles every ancillary qubit ($j \geq 1$) with the read-out qubit ($j = 0$). The circuit ends with a second data rotation $R_y(x_j)$ and a final X gate on qubit 0 before measuring Z_0.

To reveal the functional form imposed by the circuit, we propagate the measurement operator Z_0 *backwards* through each gate. At each stage only those Pauli terms that survive on the initial state $|0\rangle^{\otimes n}$ can contribute to the final expectation value. Table I lists the observable O_k after conjugation by the gates located to its right in the circuit.

TABLE I: Output tracing of the measurement operator Z_0.

Stage	Gate(s) conjugated	Observable O_k
0	–	Z_0
1	X_0 & $R_y^{(0)}(\frac{\pi}{2})$	$-X_0$
2	$\prod_{j=1}^{n-1} \mathrm{CZ}_{0j}$ & $R_y^{(i)}(\frac{\pi}{2})$, $i \geq 1$	$-\prod_{j=1}^{n-1} Z_j$
3	$R_y^{(j)}(w_j)$, $j \geq 1$	$-\prod_{j=1}^{n-1}(X_j \cos w_j + Z_j \sin w_j)$
4	H_j, $j \geq 1$	$-\prod_{j=1}^{n-1}(Z_j \cos w_j + X_j \sin w_j)$
5	$R_y^{(j)}(x_j)$, $j \geq 1$	$-\prod_{j=1}^{n-1}[Z_j \cos(w_j - x_j) + X_j \sin(w_j - x_j)]$

Because the initial state is an X-eigenstate with eigenvalue $+1$, every term containing an X_j ($j \geq 0$) has zero expectation. Consequently the only surviving contribution is

$$\langle Z_0 \rangle = -\prod_{j=1}^{n-1} \cos(w_j - x_j)$$

which manifests a multiplicative cosine kernel over the feature differences $w_j - x_j$.

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C. Training

All models are trained end-to-end on parameterized quantum circuits using the Adam optimizer over 50 epochs. For each configuration of encoding method, dimensionality reduction, feature map, and ansatz, we run three independent trials using different random seeds.

During training, each circuit is measured via the expectation value of a Pauli-Z observable on the first qubit. This scalar output is then used to infer the binary class label. We report accuracy and training time as the main evaluation metrics, along with standard deviation across seeds to capture variability in performance.

D. Dataset

We use the Amazon Cell Phones sentiment dataset extracted from the corpus provided by Kotzias et al. [21]. The dataset contains 1 000 short English-language product reviews, each labeled as either positive or negative sentiment. The samples are perfectly balanced, with 500 positive and 500 negative entries.

Each review is a single sentence. The average sentence length is approximately 10.2 words, with a minimum of 1 and a maximum of 30. This compact structure makes the dataset well-suited for quantum classification tasks, where the number of input features directly impacts the number of required qubits.

IV. Experimental Results

Experiments are done in the order mentioned in the previous section, starting with the variation of qubit number. Table II shows a clear monotonic rise in wall-clock training cost as the circuit width grows: runtime increases from roughly 280 s at $k = 2$ to 773 s at $k = 6$, a nearly three-fold jump that mirrors the exponential growth of the simulator's Hilbert-space dimension. Accuracy, however, does not follow the same trajectory. Performance is essentially random at $k = 2$ and $k = 3$ (test accuracy ≈ 0.50), climbs sharply to its peak at $k = 4$ with 0.603 ± 0.008, and then declines again as additional qubits are added, falling to 0.538 ± 0.008 at $k = 6$.

The crossover at $k = 4$ therefore marks the best trade-off between expressive capacity and resource overhead. Larger circuits appear to suffer from over-parameterisation or amplified simulation noise, eroding any accuracy gains while still incurring extra computational cost. Conversely, circuits with fewer than four qubits lack sufficient degrees of freedom to model the decision boundary. For these reasons we fix $k = 4$ in all subsequent component-wise experiments.

Table III evaluates six sentence–embedding strategies while keeping every downstream quantum component fixed to the baseline configuration identified earlier ($k = 4$, PCA, Pauli encoding, Real-Amplitude ansatz). Vocabulary-based encoders (BoW and TF-IDF) cluster near the same performance envelope as before, with test accuracies of 0.603 and 0.608, respectively. Switching to dense word-level embeddings yields mixed results where FastText under-performs all baselines on the test split (0.520) despite comparable training time, whereas Word2Vec generalises slightly better (0.605) but offers no computational advantage.

TABLE II: Baseline Accuracies and Training Time by Qubit Numbers (k)

k	Train Acc	Test Acc	Training Time (s)
2	0.514 ± 0.003	0.495 ± 0.02	**278.75** ± 9.934
3	0.542 ± 0.003	0.497 ± 0.006	390.15 ± 0.113
4	**0.621** ± 0.005	**0.603** ± 0.008	507.87 ± 0.157
5	0.607 ± 0.01	0.575 ± 0.03	625.64 ± 0.578
6	0.599 ± 0.003	0.538 ± 0.008	772.98 ± 4.435

Sentence-level transformers deliver a decisive jump in accuracy. MiniLM attains the best test score (**0.740 ± 0.009**), a 13-percentage-point gain over BoW, while MPNet records the highest training accuracy (0.746) and essentially ties MiniLM on inference (0.737 ± 0.028). Because MiniLM and MPNet outperform the other four methods, both are carried forward as the default methods in the subsequent component studies.

TABLE III: Comparison of Sentence Embedding Methods

Embedding	Train Acc	Test Acc	Training Time (s)
BoW	0.621 ± 0.005	0.603 ± 0.008	507.87 ± 0.157
TF-IDF	0.608 ± 0.018	0.608 ± 0.020	509.849 ± 11.560
FastText	0.579 ± 0.014	0.520 ± 0.017	506.288 ± 0.475
Word2Vec	0.596 ± 0.018	0.605 ± 0.035	**505.542** ± 0.962
MiniLM	0.733 ± 0.01	**0.740** ± 0.009	506.604 ± 0.189
MPNet	**0.746** ± 0.003	0.737 ± 0.028	507.525 ± 0.079

Table IV compares PCA and UMAP applied to the best performing encoders, MiniLM and MPNet, while every subsequent quantum component is kept constant. In both cases PCA outperforms UMAP by a wide margin. For MiniLM the switch to UMAP drops test accuracy from 0.740 ± 0.009 to 0.493 ± 0.055, effectively erasing the gains delivered by the transformer embedding. MPNet exhibits the same pattern: PCA yields 0.737 ± 0.028 on the test split, whereas UMAP falls to 0.535 ± 0.028.

These results indicate that, for this task and circuit width, linear variance preservation (PCA) is preferable to UMAP's non-linear manifold projection, which appears to scramble information critical to the downstream quantum classifier. Consequently, PCA is fixed as the default reducer in all remaining experiments.

Table V explores how the choice of qubit encoding, Angle and Pauli, affects performance once the pipeline has been fixed to MiniLM or MPNet embeddings and PCA dimensionality reduction. In both cases Angle encoding decisively outperforms the baseline Pauli scheme. With MiniLM the switch raises test accuracy from 0.740 to 0.797 while cutting training time by roughly 74 seconds (507.6 s to 432.8 s). The improvement is

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TABLE IV: Comparison of Dimensionality Reduction Methods

Embedding	Reducer	Train Acc	Test Acc	Training Time (s)
MiniLM	PCA	0.733 ± 0.01	**0.740** ± 0.009	**506.604** ± 0.189
MiniLM	UMAP	0.568 ± 0.021	0.493 ± 0.055	528.153 ± 22.250
MPNet	PCA	**0.746** ± 0.003	0.737 ± 0.028	507.525 ± 0.079
MPNet	UMAP	0.571 ± 0.037	0.535 ± 0.028	515.624 ± 0.276

even more pronounced for MPNet, where test accuracy climbs from 0.737 to 0.870 and runtime drops by about 86 seconds.

The gains stem from Angle encoding's lower circuit depth and the absence of entangling rotations, which reduces noise accumulation and accelerates simulation while still providing sufficient feature expressiveness. Pauli encoding, by contrast, embeds each feature as a full Pauli rotation, introducing additional gates that appear to over-parameterise the circuit and inflate optimisation cost. Given these consistent wins in both accuracy and training time, Angle encoding is selected as the default feature map for the subsequent ansatz comparison.

TABLE V: Comparison of Feature Encoding Strategies

Embedding	Encoding	Train Acc	Test Acc	Training Time (s)
MiniLM	Angle	0.802 ± 0.002	0.797 ± 0.003	432.830 ± 19.621
MiniLM	Pauli	0.733 ± 0.01	0.740 ± 0.009	506.604 ± 0.189
MPNet	Angle	**0.843** ± 0.005	**0.870** ± 0.005	**421.964** ± 0.323
MPNet	Pauli	0.746 ± 0.003	0.737 ± 0.028	507.525 ± 0.079

Table VI completes the study by swapping the variational ansatz while holding all preceding stages fixed (MiniLM or MPNet embeddings, PCA, Angle encoding, $k = 4$). The Real-Amplitude ansatz remains a strong baseline, delivering the single highest test accuracy (0.870 with MPNet) at moderate cost (422 s), whereas the Strongly Entangling Layer (SEL) is consistently the slowest option (604 s) and offers no compensating gain in generalisation.

The custom cosine-interference circuit offers the best trade-off between accuracy and training time. With MiniLM, it improves the test score from 0.797 to 0.812 while reducing runtime from 433 s to 287 s. When paired with MPNet, it achieves 0.865 test accuracy, just 0.005 points behind Real Amplitude ansatz, while completing training in just 277 s, over 35% faster than the best-performing Real-Amplitude ansatz (0.870 at 422 s). These results indicate that deeper entanglement, as used in SEL, does not necessarily yield better performance in this task. Instead, a carefully tailored shallow circuit can approach peak accuracy at significantly lower computational cost.

From the best-performing model, we examine a selection of both correctly and incorrectly classified examples, as shown in Table VII. These representative cases are analyzed qualitatively to better understand the model's decision patterns, strengths, and potential sources of error.

TABLE VI: Comparison of Ansatz Choices

Embedding	Ansatz	Train Acc	Test Acc	Training Time (s)
MiniLM	Real Ampl.	0.802 ± 0.002	0.797 ± 0.003	432.832 ± 19.621
MiniLM	Custom	0.807 ± 0.007	0.812 ± 0.003	287.356 ± 9.925
MiniLM	SEL	0.796 ± 0.002	0.792 ± 0.006	603.570 ± 0.061
MPNet	Real Ampl.	0.843 ± 0.005	**0.870** ± 0.005	421.964 ± 0.323
MPNet	Custom	0.852 ± 0.004	0.865 ± 0.005	**277.282** ± 0.103
MPNet	SEL	**0.856** ± 0.002	0.860 ± 0.005	604.695 ± 1.003

TABLE VII: Representative predictions by the MPNet–PCA–Angle-Real Amplitude model

Case type	Sentence fragment	True	Pred.
TP	I was skeptical at first, but this phone proved me wrong	+	+
TP	Even though it's refurbished, everything works perfectly	+	+
TN	Don't be fooled by the specs—this device is painfully slow	–	–
TN	The design is nice, but that's the only good thing I can say	–	–
FP	No buyers remorse on this one	–	+
FP	Save your money … I've had this item for months now	–	+
FN	Only had this a month but it's worked flawlessly so far	+	–
FN	My headset works just *peachy keen*	+	–

Table VII presents a selection of representative predictions made by the MPNet-PCA$_4$–Angle-Real Amplitude model, including both correct and incorrect classifications of nuanced cases. The true positive (TP) and true negative (TN) examples show that the model is capable of capturing implied sentiment and handling contrastive or concessive structures. For example, it correctly identifies positivity in *"I was skeptical at first, but this phone proved me wrong"*, and detects criticism in the misleading praise of *"Don't be fooled by the specs—this device is painfully slow"*. These cases suggest that the model is not merely matching keywords, but is able to process some forms of implicit sentiment when the linguistic cues are sufficiently clear or follow familiar patterns.

However, the false positive (FP) and false negative (FN) examples highlight areas where the model remains vulnerable. FP cases such as *"No buyers remorse on this one"* reveal difficulties with negation, while sarcastic warnings like *"Save your money…"* are misclassified due to their superficially positive tone. FN cases, on the other hand, suggest that the model may underperform when praise is hedged or expressed in informal language, as in *"My headset works just peachy keen"*, or when uncertain temporal framing is present. These errors indicate a sensitivity to surface-level features that occasionally

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overrides broader context and pragmatic cues.

TABLE VIII: Comparison of Classical and Quantum Architecture Choices (MPNet–PCA–Angle)

Embedding	Arch.	Train Acc	Test Acc	Training Time (s)
MPNet	Classical	**0.875** $\pm$ 0.003	0.853 $\pm$ 0.006	**5.081** $\pm$ 0.829
MPNet	Real Ampl.	0.843 $\pm$ 0.005	**0.870** $\pm$ 0.005	421.964 $\pm$ 0.323
MPNet	Custom	0.852 $\pm$ 0.004	0.865 $\pm$ 0.005	277.282 $\pm$ 0.103
MPNet	SEL	0.856 $\pm$ 0.002	0.860 $\pm$ 0.005	604.695 $\pm$ 1.003

Table VIII compares the best-performing quantum configurations with a classical baseline under the same embedding and preprocessing pipeline (MPNet–PCA–Angle). The classical model takes the reduced embedding as input, passes it through one hidden layer with 64 units, and outputs a score indicating positive or negative sentiment. This model achieves strong performance with 85.3% test accuracy at negligible training cost. Quantum circuits, while substantially slower due to simulation overhead, are competitive in accuracy, with the Real-Amplitude ansatz reaches the highest score of 87.0%, slightly surpassing the classical baseline, while the custom cosine-interference circuit offers a similar 86.5% accuracy with a 35% reduction in runtime relative to the Real-Amplitude ansatz. These results illustrate that, although classical models remain superior in efficiency on today's hardware, carefully designed quantum pipelines can approach or even match classical performance levels in simulation, highlighting their potential as NISQ-era methods mature.

This study demonstrates that carefully selecting each component of a QNLP sentiment classification pipeline can lead to significant accuracy gains and reduced training time. Fixing the circuit width at $k = 4$ offers a strong balance between model expressiveness and runtime cost. Transformer-based sentence embeddings (MiniLM, MPNet) elevate test accuracy from around 0.60 to around 0.74, and PCA preserves these gains while keeping the feature size manageable. Compared to Pauli encoding, angle encoding improves both speed and accuracy in this setting. Under the final configuration of 4 qubits, MPNet embeddings, PCA dimension reduction method, and angle encoding, the Real-Amplitude ansatz achieves the highest test accuracy of 87%. Evaluated under the same setup, the custom cosine-interference ansatz closely follows with 86.5% accuracy while reducing training time by more than 35% (277 s vs. 432 s).

This study was conducted entirely on simulated, noise-free back-ends. A natural next step is to deploy the best-performing configurations on real quantum hardware and incorporate error mitigation techniques to counteract decoherence and gate noise. Future experiments should also extend to larger and more diverse NLP datasets to assess how well the pipeline generalises across domains. In terms of methodology, further improvements could be explored through more expressive encoding schemes such as trainable feature maps or data re-uploading techniques that may enhance performance when paired with lightweight ansatz.

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